KAIZEN MATHS

Lesson starter

Motion Graphs and Constant Acceleration

Constant Acceleration

Name _____

SA1

An object has initial velocity 3 m/s and accelerates uniformly at 2 m/s² for 4 s. Find its final velocity and distance travelled.

MEC-50A8-L3 · SA

Show your working

KAIZEN MATHS

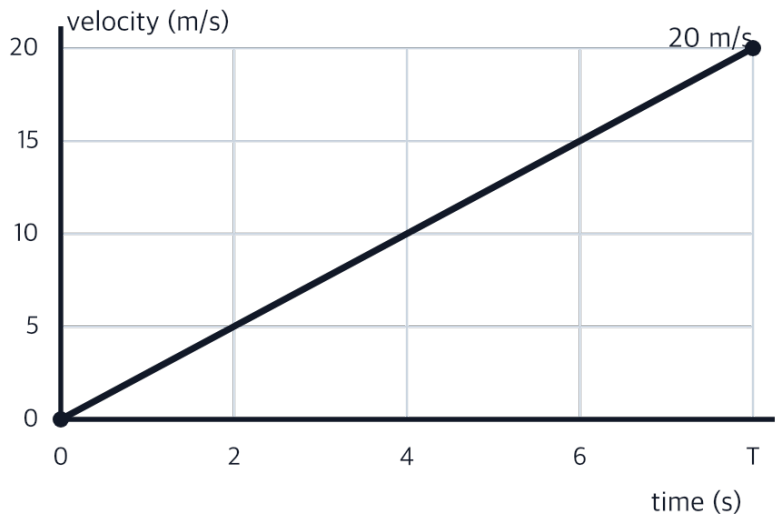
Lesson starter

Motion Graphs and Constant Acceleration

Constant Acceleration

Name _____

SB1



The triangular velocity-time graph has maximum velocity 20 m/s. The distance travelled is 80 m. Find the value of T.

MEC-50A8-L3 · SB

Show your working

KAIZEN MATHS

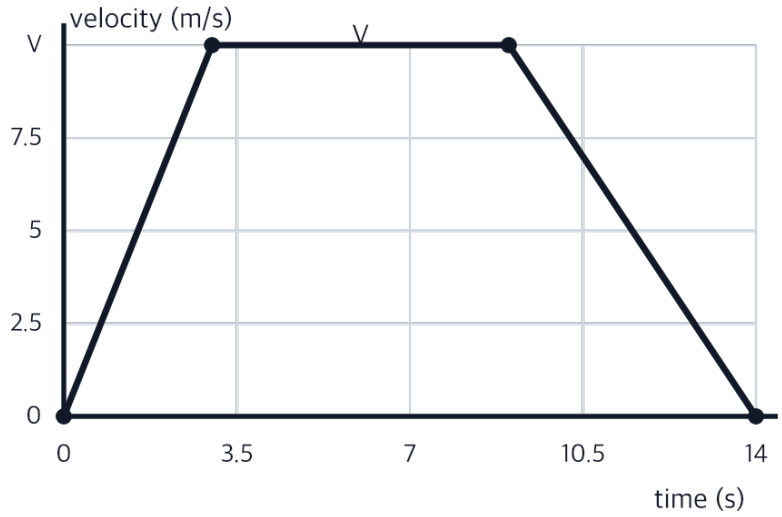
Lesson starter

Motion Graphs and Constant Acceleration

Constant Acceleration

Name _____

SC1



The velocity-time graph shows a journey lasting 14 s. The total distance travelled is 100 m. Find the value of V.

MEC-50A8-L3 · SC

Show your working

KAIZEN MATHS

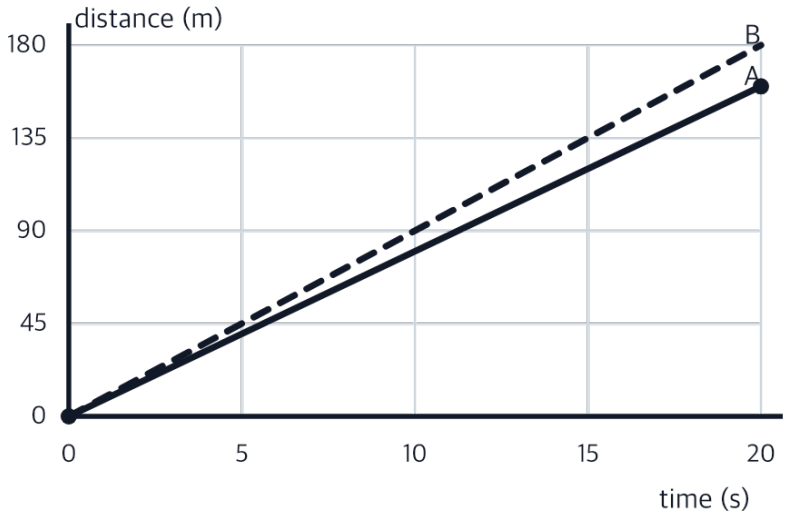
Lesson starter

Motion Graphs and Constant Acceleration

Constant Acceleration

Name _____

SD1



The solid line shows journey A and the dashed line shows journey B. Find the average speed of each journey and state which is faster.

MEC-50A8-L3 · SD

Show your working

KAIZEN MATHS

Exit ticket

Motion Graphs and Constant Acceleration

Constant Acceleration

Name

EA1

An object has initial velocity 3 m/s and accelerates uniformly at 2 m/s² for 8 s. Find its final velocity and distance travelled.

Show your working

KAIZEN MATHS

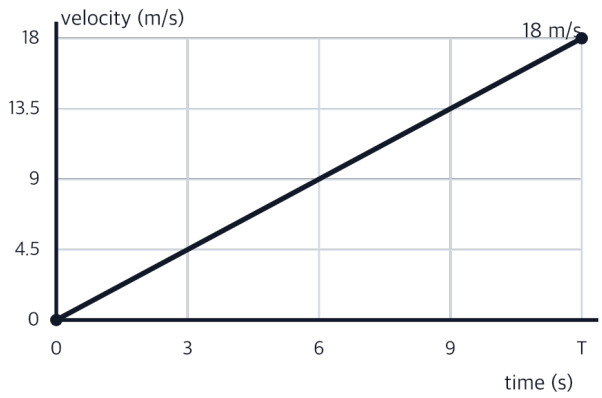
Exit ticket

Motion Graphs and Constant Acceleration

Constant Acceleration

Name

EB1



The triangular velocity-time graph has maximum velocity 18 m/s. The distance travelled is 108 m. Find the value of T.

Show your working

KAIZEN MATHS

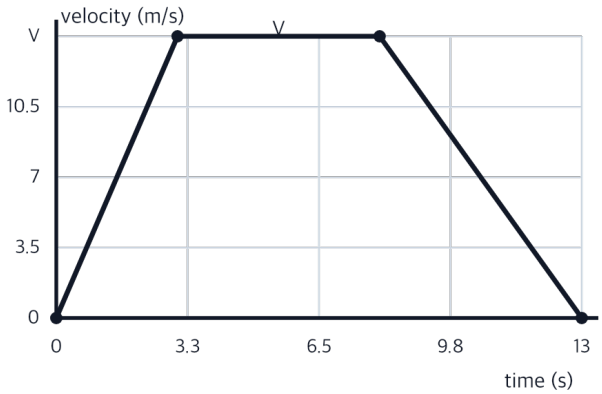
Exit ticket

Motion Graphs and Constant Acceleration

Constant Acceleration

Name

EC1



The velocity-time graph shows a journey lasting 13 s. The total distance travelled is 126 m. Find the value of V.

Show your working

KAIZEN MATHS

Exit ticket

Motion Graphs and Constant Acceleration

Constant Acceleration

Name

EA1

An object has initial velocity 3 m/s and accelerates uniformly at 2 m/s² for 8 s. Find its final velocity and distance travelled.

Show your working

Teacher answers & working

Motion Graphs and Constant Acceleration · Constant Acceleration

MEC-50A8-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

An object has initial velocity 3 m/s and accelerates uniformly at 2 m/s² for 4 s. Find its final velocity and distance travelled.

Answer:

$v = 11$ m/s, $s = 28$ m

Working:

List the known values.

$u = 3$ m/s, $a = 2$ m/s^2, $t = 4$ s

Use v = u + at to find final velocity.

$v = 3 + 2 \times 4 = 11$

Use s = ut + ½at² to find distance.

$s = 3 \times 4 + ½ \times 2 \times 4^2$

$s = 12 + 16 = 28$

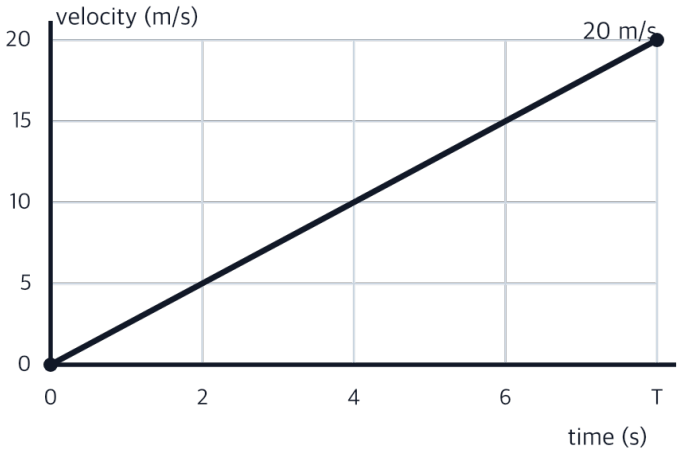
Final velocity = 11 m/s and distance = 28 m.

Teacher answers & working

Motion Graphs and Constant Acceleration · Constant Acceleration

MEC-50A8-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



The triangular velocity-time graph has maximum velocity 20 m/s. The distance travelled is 80 m. Find the value of T.

Answer:

$T = 8$ s

Working:

Distance travelled is the area under the velocity-time graph.

The area is a triangle.

area = ½ × base × height

$80 = ½ \times T \times 20$

Rearrange to find T.

$T = (80 \times 2) \div 20$

$T = 8$

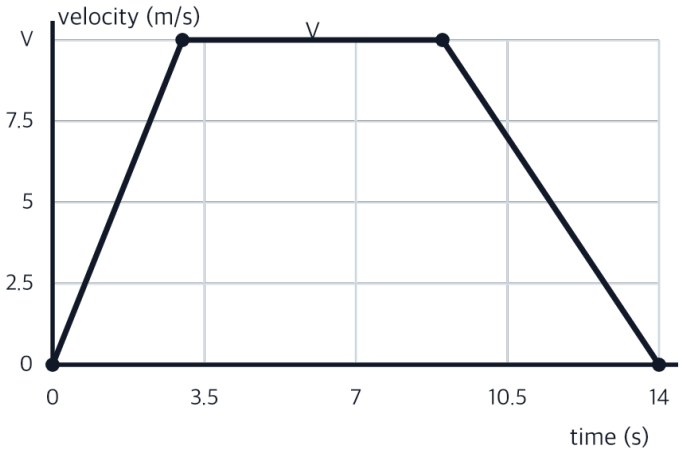
$T = 8$ s.

Teacher answers & working

Motion Graphs and Constant Acceleration · Constant Acceleration

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SC1



The velocity-time graph shows a journey lasting 14 s. The total distance travelled is 100 m. Find the value of V.

Answer:

$V = 10\ m/s$

Working:

Distance travelled is the area under the graph.

Split the graph into a triangle, a rectangle, and a triangle.

area = ½ × 3 × V + (9 - 3) × V + ½ × (14 - 9) × V

area = 1.5V + 6V + 2.5V

$100 = 10V$

Divide by the coefficient of V.

$V = 100 \div 10 = 10$

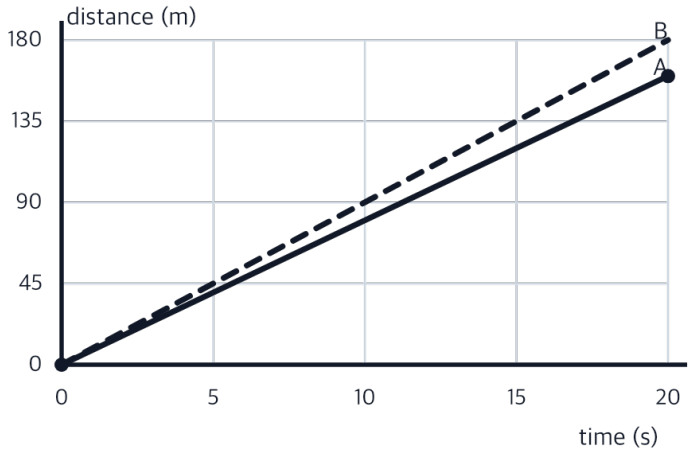
$V = 10\ m/s.$

Teacher answers & working

Motion Graphs and Constant Acceleration · Constant Acceleration

MEC-50A8-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



The solid line shows journey A and the dashed line shows journey B. Find the average speed of each journey and state which is faster.

Answer:

A = 8 m/s, B = 9 m/s; B is faster

Working:

On a distance-time graph, average speed is gradient from start to finish.

journey A speed = 160 ÷ 20 = 8 m/s

journey B speed = 180 ÷ 20 = 9 m/s

Compare the gradients.

Journey B is faster because its gradient is greater.

EA1

An object has initial velocity 3 m/s and accelerates uniformly at 2 m/s² for 8 s. Find its final velocity and distance travelled.

Answer:

$v = 19$ m/s, $s = 88$ m

Working:

List the known values.

$u = 3$ m/s, $a = 2$ m/s^2, $t = 8$ s

Use v = u + at to find final velocity.

$v = 3 + 2 \times 8 = 19$

Use s = ut + ½at² to find distance.

$s = 3 \times 8 + ½ \times 2 \times 8^2$

$s = 24 + 64 = 88$

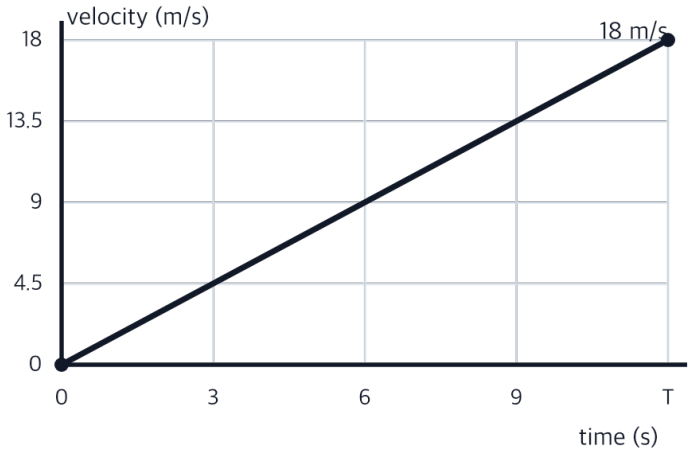
Final velocity = 19 m/s and distance = 88 m.

Teacher answers & working

Motion Graphs and Constant Acceleration · Constant Acceleration

MEC-50A8-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



The triangular velocity-time graph has maximum velocity 18 m/s. The distance travelled is 108 m. Find the value of T.

Answer:

$T = 12$ s

Working:

Distance travelled is the area under the velocity-time graph.

The area is a triangle.

area = ½ × base × height

$108 = ½ \times T \times 18$

Rearrange to find T.

$T = (108 \times 2) \div 18$

$T = 12$

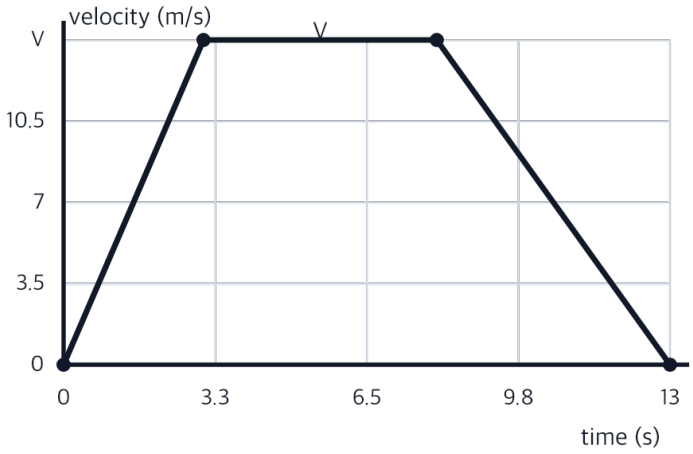
$T = 12$ s.

Teacher answers & working

Motion Graphs and Constant Acceleration · Constant Acceleration

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EC1



The velocity-time graph shows a journey lasting 13 s. The total distance travelled is 126 m. Find the value of V.

Answer:

$V = 14\ m/s$

Working:

Distance travelled is the area under the graph.

Split the graph into a triangle, a rectangle, and a triangle.

area = ½ × 3 × V + (8 - 3) × V + ½ × (13 - 8) × V

area = 1.5V + 5V + 2.5V

$126 = 9V$

Divide by the coefficient of V.

$V = 126 \div 9 = 14$

$V = 14\ m/s.$